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Probability

IMPORTANT FACTS AND FORMULAE

- I. Experiment:** An operation which can produce some well-defined outcomes is called an experiment.
II. Random Experiment: An experiment in which all possible outcomes are known and the exact output cannot be predicted in advance, is called a random experiment.

Examples of Performing a Random Experiment:

- (i) Rolling an unbiased dice
- (ii) Tossing a fair coin
- (iii) Drawing a card from a pack of well-shuffled cards
- (iv) Picking up a ball of certain colour from a bag containing balls of different colours

Details :

- (i) When we throw a coin, then either a Head (H) or a Tail (T) appears.
- (ii) A dice is a solid cube, having 6 faces, marked 1, 2, 3, 4, 5, 6 respectively.
When we throw a die, the outcome is the number that appears on its upper face.
- (iii) A pack of cards has 52 cards.
It has 13 cards of each suit, namely **Spades, Clubs, Hearts** and **Diamonds**.
Cards of spades and clubs are **black cards**.
Cards of hearts and diamonds are **red cards**.
There are 4 honours of each suit.
These are **Aces, Kings, Queens** and **Jacks**.
These are called **face cards**.

- III. Sample Space:** When we perform an experiment, then the set S of all possible outcomes is called the **Sample Space**.

Examples of Sample Spaces:

- (i) In tossing a coin, $S = \{H, T\}$.
- (ii) If two coins are tossed, then $S = \{HH, HT, TH, TT\}$.
- (iii) In rolling a dice, we have, $S = \{1, 2, 3, 4, 5, 6\}$.

- IV. Event :** Any subset of a sample space is called an event.

V. Probability of Occurrence of an Event:

Let S be the sample space and let E be an event. Then, $E \subseteq S$.

$$\therefore P(E) = \frac{n(E)}{n(S)}$$

VI. Results on Probability:

- (i) $P(S) = 1$ (ii) $0 \leq P(E) \leq 1$ (iii) $P(\phi) = 0$
- (iv) For any events A and B, we have :
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- (v) If $\bar{A}$ denotes (not-A), then $P(\bar{A}) = 1 - P(A)$.

SOLVED EXAMPLES

Ex. 1. In a throw of a coin, find the probability of getting a head.

Sol. Here $S = \{H, T\}$ and $E = \{H\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{1}{2}$$

Ex. 2. Two unbiased coins are tossed. What is the probability of getting at most one head ?

Sol. Here $S = \{HH, HT, TH, TT\}$.

Let E = event of getting at most one head.

$\therefore E = \{TT, HT, TH\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}.$$

Ex. 3. An unbiased die is tossed. Find the probability of getting a multiple of 3.

Sol. Here $S = \{1, 2, 3, 4, 5, 6\}$.

Let E be the event of getting a multiple of 3.

Then, $E = \{3, 6\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

Ex. 4. In a simultaneous throw of a pair of dice, find the probability of getting a total more than 7.

Sol. Here, $n(S) = (6 \times 6) = 36$.

Let E = Event of getting a total more than 7

$= \{(2, 6), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6), (5, 3), (5, 4), (5, 5), (5, 6), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}.$$

Ex. 5. A bag contains 6 white and 4 black balls. Two balls are drawn at random. Find the probability that they are of the same colour.

Sol. Let S be the sample space. Then,

$$n(S) = \text{Number of ways of drawing 2 balls out of } (6 + 4) = {}^{10}C_2 = \frac{(10 \times 9)}{(2 \times 1)} = 45.$$

Let E = Event of getting both balls of the same colour. Then,

$n(E)$ = Number of ways of drawing (2 balls out of 6) or (2 balls out of 4)

$$= ({}^6C_2 + {}^4C_2) = \frac{(6 \times 5)}{(2 \times 1)} + \frac{(4 \times 3)}{(2 \times 1)} = (15 + 6) = 21.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{21}{45} = \frac{7}{15}.$$

Ex. 6. Two dice are thrown together. What is the probability that the sum of the numbers on the two faces is divisible by 4 or 6 ?

Sol. Clearly, $n(S) = 6 \times 6 = 36$.

Let E be the event that the sum of the numbers on the two faces is divisible by 4 or 6. Then

$E = \{(1, 3), (1, 5), (2, 2), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (5, 1), (5, 3), (6, 2), (6, 6)\}$

$\therefore n(E) = 14$.

$$\text{Hence, } P(E) = \frac{n(E)}{n(S)} = \frac{14}{36} = \frac{7}{18}.$$

Ex. 7. Two cards are drawn at random from a pack of 52 cards. What is the probability that either both are black or both are queens ?

Sol. We have $n(S) = {}^{52}C_2 = \frac{(52 \times 51)}{(2 \times 1)} = 1326$.

Let A = event of getting both black cards;

B = event of getting both queens.

$\therefore A \cap B$ = event of getting queens of black cards.

$$\therefore n(A) = {}^{26}C_2 = \frac{(26 \times 25)}{(2 \times 1)} = 325, n(B) = {}^4C_2 = \frac{(4 \times 3)}{(2 \times 1)} = 6 \text{ and } n(A \cap B) = {}^2C_2 = 1.$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{325}{1326}; P(B) = \frac{n(B)}{n(S)} = \frac{6}{1326} \text{ and } P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{1326}.$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) = \left(\frac{325}{1326} + \frac{6}{1326} - \frac{1}{1326} \right) = \frac{330}{1326} = \frac{55}{221}.$$

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. In a simultaneous throw of two coins, the probability of getting at least one head is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

2. Three unbiased coins are tossed. What is the probability of getting at least 2 heads?

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{3}$ (d) $\frac{1}{8}$

3. Three unbiased coins are tossed. What is the probability of getting at most two heads?

(a) $\frac{3}{4}$ (b) $\frac{1}{4}$
(c) $\frac{3}{8}$ (d) $\frac{7}{8}$

4. In a single throw of a die, what is the probability of getting a number greater than 4?

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{4}$

5. In a simultaneous throw of two dice, what is the probability of getting a total of 7?

(a) $\frac{1}{6}$ (b) $\frac{1}{4}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

6. What is the probability of getting a sum 9 from two throws of a dice?

(a) $\frac{1}{6}$ (b) $\frac{1}{8}$
(c) $\frac{1}{9}$ (d) $\frac{1}{12}$

7. In a simultaneous throw of two dice, what is the probability of getting a doublet?

(a) $\frac{1}{6}$ (b) $\frac{1}{4}$
(c) $\frac{2}{3}$ (d) $\frac{3}{7}$

8. In a simultaneous throw of two dice, what is the probability of getting a total of 10 or 11?

(a) $\frac{1}{4}$ (b) $\frac{1}{6}$
(c) $\frac{7}{12}$ (d) $\frac{5}{36}$

9. Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even?

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$
(c) $\frac{3}{8}$ (d) $\frac{5}{16}$

10. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn bears a number which is a multiple of 3?

(a) $\frac{3}{10}$ (b) $\frac{3}{20}$
(c) $\frac{2}{5}$ (d) $\frac{1}{2}$

11. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

(a) $\frac{1}{2}$ (b) $\frac{2}{5}$
(c) $\frac{8}{15}$ (d) $\frac{9}{20}$

12. In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?

(a) $\frac{1}{10}$ (b) $\frac{2}{5}$
(c) $\frac{2}{7}$ (d) $\frac{5}{7}$

13. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card?

(a) $\frac{1}{13}$ (b) $\frac{4}{13}$
(c) $\frac{1}{4}$ (d) $\frac{9}{52}$

14. A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is

(a) $\frac{1}{13}$ (b) $\frac{2}{13}$

- (c) $\frac{1}{26}$ (d) $\frac{1}{52}$
15. One card is drawn from a pack of 52 cards. What is the probability that the card drawn is either a red card or a king ?
- (a) $\frac{1}{2}$ (b) $\frac{6}{13}$
- (c) $\frac{7}{13}$ (d) $\frac{27}{52}$
16. From a pack of 52 cards, one card is drawn at random. What is the probability that the card drawn is a ten or a spade ?
- (a) $\frac{4}{13}$ (b) $\frac{1}{4}$
- (c) $\frac{1}{13}$ (d) $\frac{1}{26}$
17. The probability that a card drawn from a pack of 52 cards will be a diamond or a king, is
- (a) $\frac{2}{13}$ (b) $\frac{4}{13}$
- (c) $\frac{1}{13}$ (d) $\frac{1}{52}$
18. From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings ?
- (a) $\frac{1}{15}$ (b) $\frac{25}{57}$
- (c) $\frac{35}{256}$ (d) $\frac{1}{221}$
19. Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart, is
- (a) $\frac{3}{20}$ (b) $\frac{29}{34}$
- (c) $\frac{47}{100}$ (d) $\frac{13}{102}$
20. Two cards are drawn from a pack of 52 cards. The probability that either both are red or both are kings, is
- (a) $\frac{7}{13}$ (b) $\frac{3}{26}$
- (c) $\frac{63}{221}$ (d) $\frac{55}{221}$
21. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white ?
- (a) $\frac{3}{4}$ (b) $\frac{4}{7}$
- (c) $\frac{1}{8}$ (d) $\frac{3}{7}$
- (e) None of these
22. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green ?
- (a) $\frac{2}{3}$ (b) $\frac{3}{4}$
- (c) $\frac{7}{19}$ (d) $\frac{8}{21}$
- (e) $\frac{9}{21}$
23. A box contains 4 red, 5 green and 6 white balls. A ball is drawn at random from the box. What is the probability that the ball drawn is either red or green?
- (a) $\frac{2}{5}$ (b) $\frac{3}{5}$
- (c) $\frac{1}{5}$ (d) $\frac{7}{15}$
- (e) None of these
24. A basket contains 4 red, 5 blue and 3 green marbles. If 2 marbles are drawn at random from the basket, what is the probability that both are red?
- (S.B.I. P.O., 2010)
- (a) $\frac{3}{7}$ (b) $\frac{1}{2}$
- (c) $\frac{1}{11}$ (d) $\frac{1}{6}$
- (e) None of these
25. An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If two marbles are drawn at random from the urn, what is the probability that both are red?
- (S.B.I. P.O., 2010)
- (a) $\frac{1}{6}$ (b) $\frac{1}{7}$
- (c) $\frac{2}{15}$ (d) $\frac{2}{5}$
- (e) None of these
26. A basket contains 6 blue, 2 red, 4 green and 3 yellow balls. If three balls are picked up at random, what is the probability that none is yellow? (Bank P.O., 2009)
- (a) $\frac{3}{455}$ (b) $\frac{1}{5}$
- (c) $\frac{4}{5}$ (d) $\frac{44}{91}$
- (e) None of these

27. An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If three marbles are picked up at random, what is the probability that 2 are blue and 1 is yellow? (S.B.I. P.O., 2010)
- (a) $\frac{3}{91}$ (b) $\frac{1}{5}$
 (c) $\frac{18}{455}$ (d) $\frac{7}{15}$
 (e) None of these
28. An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If four marbles are picked up at random, what is the probability that 1 is green, 2 are blue and 1 is red? (S.B.I. P.O., 2011)
- (a) $\frac{13}{35}$ (b) $\frac{24}{455}$
 (c) $\frac{11}{15}$ (d) $\frac{1}{13}$
 (e) None of these
29. An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If two marbles are picked up at random, what is the probability that either both are green or both are yellow? (Bank P.O., 2010)
- (a) $\frac{5}{91}$ (b) $\frac{1}{35}$
 (c) $\frac{1}{3}$ (d) $\frac{4}{105}$
 (e) None of these
30. A basket contains 6 blue, 2 red, 4 green and 3 yellow balls. If four balls are picked up at random, what is the probability that 2 are red and 2 are green? (Bank P.O., 2009)
- (a) $\frac{4}{15}$ (b) $\frac{5}{27}$
 (c) $\frac{1}{3}$ (d) $\frac{2}{455}$
 (e) None of these
31. A basket contains 4 red, 5 blue and 3 green marbles. If three marbles are picked up at random what is the probability that at least one is blue? (S.B.I. P.O., 2010)
- (a) $\frac{7}{12}$ (b) $\frac{37}{44}$
 (c) $\frac{5}{12}$ (d) $\frac{7}{44}$
 (e) None of these
32. An urn contains 6 red, 4 blue, 2 green and 3 yellow marbles. If 4 marbles are picked up at random, what is the probability that at least one of them is blue? (S.B.I. P.O., 2010)
- (a) $\frac{4}{15}$ (b) $\frac{69}{91}$
 (c) $\frac{11}{15}$ (d) $\frac{22}{91}$
 (e) None of these
33. A basket contains 6 blue, 2 red, 4 green and 3 yellow balls. If 5 balls are picked up at random, what is the probability that at least one is blue? (Bank P.O., 2009)
- (a) $\frac{137}{143}$ (b) $\frac{18}{455}$
 (c) $\frac{9}{91}$ (d) $\frac{2}{5}$
 (e) None of these
34. An urn contains 2 red, 3 green and 2 blue balls. If 2 balls are drawn at random, find the probability that no ball is blue. (Railways, 2006)
- (a) $\frac{5}{7}$ (b) $\frac{10}{21}$
 (c) $\frac{2}{7}$ (d) $\frac{11}{21}$
 (e) None of these
35. A box contains 10 black and 10 white balls. What is the probability of drawing 2 balls of the same colour?
- (a) $\frac{9}{19}$ (b) $\frac{9}{38}$
 (c) $\frac{10}{19}$ (d) $\frac{5}{19}$
 (e) None of these
36. A box contains 20 electric bulbs, out of which 4 are defective. Two balls are chosen at random from this box. The probability that at least one of them is defective, is
- (a) $\frac{4}{19}$ (b) $\frac{7}{19}$
 (c) $\frac{12}{19}$ (d) $\frac{21}{95}$
 (e) None of these
37. In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that the selected students are 2 boys and 1 girl, is:
- (a) $\frac{21}{46}$ (b) $\frac{25}{117}$
 (c) $\frac{1}{50}$ (d) $\frac{3}{25}$
 (e) None of these
38. Four persons are chosen at random from a group of 3 men, 2 women and 4 children. The chance that exactly 2 of them are children, is
- (a) $\frac{1}{9}$ (b) $\frac{1}{5}$

- (c) $\frac{1}{12}$ (d) $\frac{10}{21}$
 (e) None of these
39. Two dice are tossed. The probability that the total score is a prime number is
 (a) $\frac{1}{6}$ (b) $\frac{1}{2}$
 (c) $\frac{5}{12}$ (d) $\frac{7}{9}$
 (e) None of these
40. In a class, 30% of the students offered English, 20% offered Hindi and 10% offered both. If a student is selected at random, what is the probability that he has offered English or Hindi ?
 (a) $\frac{2}{5}$ (b) $\frac{3}{5}$
 (c) $\frac{3}{4}$ (d) $\frac{3}{10}$
 (e) None of these
41. A man and his wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $\frac{1}{7}$ and the probability of wife's selection is $\frac{1}{5}$. What is the probability that only one of them is selected ?
 (a) $\frac{4}{5}$ (b) $\frac{2}{7}$
 (c) $\frac{4}{7}$ (d) $\frac{8}{15}$
 (e) None of these
42. A speaks truth in 75% cases and B in 80% of the cases. In what percentage of cases are they likely to contradict each other, in narrating the same incident?
 (a) 5% (b) 15%
 (c) 35% (d) 45%
 (e) None of these
43. A speaks truth in 60% cases and B speaks truth in 70% cases. The probability that they will say the same thing while describing a single event, is
 (a) 0.54 (b) 0.56
 (c) 0.68 (d) 0.94
 (e) None of these
44. A committee of 3 members is to be selected out of 3 men and 2 women. What is the probability that the committee has at least 1 woman? (Bank P.O., 2008)
 (a) $\frac{1}{10}$ (b) $\frac{9}{20}$
 (c) $\frac{1}{20}$ (d) $\frac{9}{10}$
 (e) None of these
45. A bag contains 3 blue, 2 green and 5 red balls. If four balls are picked at random, what is the probability that two are green and two are blue?
 [DMRC—Customer Relationship Assistant (CRA) Exam, 2016]
 (a) $\frac{1}{18}$ (b) $\frac{1}{70}$
 (c) $\frac{3}{5}$ (d) $\frac{1}{2}$
 (e) None of these
46. Dev can hit a target 3 times in 6 shots Pawan can hit the target 2 times in 6 shots and Lakhan can hit the target 4 times in 4 shots. What is the probability that at least 2 shorts hit the target
 [DMRC—Customer Relationship Assistant (CRA) Exam, 2016]
 (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{2}$ (d) None of these
47. A bag contains 10 mangoes out of which 4 are taken out together. If one of them is found to be good, the probability that other is also good is
 [DMRC—Train Operator (Station Controller) Exam, 2016]
 (a) $\frac{1}{3}$ (b) $\frac{8}{15}$
 (c) $\frac{5}{18}$ (d) $\frac{2}{3}$
48. A bag contains 4 red, 5 yellow and 6 pink balls. Two balls are drawn at random. What is the probability that none of the balls drawn are yellow in colour?
 [IBPS—Bank PO/MT (Pre.) Exam, 2015]
 (a) $\frac{1}{7}$ (b) $\frac{3}{7}$
 (c) $\frac{2}{7}$ (d) $\frac{5}{14}$ (e) $\frac{9}{14}$
49. A bag contains 6 red balls 11 yellow balls and 5 pink balls. If two balls are drawn at random from the bag. One after another what is the probability that the first ball is red and second ball is yellow.
 [IBPS—Bank PO (Pre.) Exam, 2015]
 (a) $\frac{1}{14}$ (b) $\frac{2}{7}$
 (c) $\frac{5}{7}$ (d) $\frac{3}{14}$
 (e) None of these
50. A bag contains 4 red balls, 6 blue balls and 8 pink balls. One ball is drawn at random and replace with 3 pink balls. A probability that the first ball drawn was either red or blue in colour and the second ball drawn was pink in colour?
 [CET—(Maharashtra (MBA) Exam, 2016]
 (a) 12/21 (b) 13/17
 (c) 11/30 (d) 13/18
 (e) None of these

ANSWERS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (b) | 3. (d) | 4. (b) | 5. (a) | 6. (c) | 7. (a) | 8. (d) | 9. (b) | 10. (a) |
| 11. (d) | 12. (c) | 13. (b) | 14. (c) | 15. (c) | 16. (a) | 17. (b) | 18. (d) | 19. (d) | 20. (d) |
| 21. (b) | 22. (d) | 23. (b) | 24. (c) | 25. (b) | 26. (d) | 27. (c) | 28. (b) | 29. (d) | 30. (d) |
| 31. (b) | 32. (b) | 33. (a) | 34. (b) | 35. (a) | 36. (b) | 37. (a) | 38. (d) | 39. (c) | 40. (a) |
| 41. (b) | 42. (c) | 43. (a) | 44. (d) | 45. (b) | 46. (a) | 47. (a) | 48. (b) | 49. (b) | 50. (e) |

SOLUTIONS

1. Here $S = \{HH, HT, TH, TT\}$.

Let $E =$ event of getting at least one head $= \{HT, TH, HH\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}.$$

2. Here $S = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$.

Let $E =$ event of getting at least two heads $= \{THH, HTH, HHT, HHH\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{8} = \frac{1}{2}.$$

3. Here $S = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$.

Let $E =$ event of getting at most two heads.

Then, $E = \{TTT, TTH, THT, HTT, THH, HTH, HHT\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{7}{8}.$$

4. When a die is thrown, we have $S = \{1, 2, 3, 4, 5, 6\}$.

Let $E =$ event of getting a number greater than 4 $= \{5, 6\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

5. We know that in a simultaneous throw of two dice, $n(S) = 6 \times 6 = 36$.

Let $E =$ event of getting a total of 7

$= \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}.$$

6. In two throws of a die, $n(S) = (6 \times 6) = 36$.

Let $E =$ event of getting a sum 9

$= \{(3, 6), (4, 5), (5, 4), (6, 3)\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}.$$

7. In a simultaneous throw of two dice, $n(S) = (6 \times 6) = 36$.

Let $E =$ event of getting a doublet

$= \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}.$$

8. In a simultaneous throw of two dice, we have $n(S) = (6 \times 6) = 36$.

Let $E =$ event of getting a total of 10 or 11

$= \{(4, 6), (5, 5), (6, 4), (5, 6), (6, 5)\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{5}{36}.$$

9. In a simultaneous throw of two dice, we have $n(S) = (6 \times 6) = 36$.

Let $E =$ event of getting two numbers whose product is even.

Then, $E = \{(1, 2), (1, 4), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 4), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 4), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$.

$\therefore n(E) = 27$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{27}{36} = \frac{3}{4}.$$

10. Here, $S = \{1, 2, 3, 4, \dots, 19, 20\}$.

Let $E =$ event of getting a multiple of 3 $= \{3, 6, 9, 12, 15, 18\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{20} = \frac{3}{10}.$$

11. Here, $S = \{1, 2, 3, 4, \dots, 19, 20\}$.

Let $E =$ event of getting a multiple of 3 or 5 $= \{3, 6, 9, 12, 15, 18, 5, 10, 20\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{9}{20}.$$

12. $P(\text{getting a prize}) = \frac{10}{(10+25)} = \frac{10}{35} = \frac{2}{7}.$

13. Clearly, there are 52 cards, out of which there are 16 face cards.

$$\therefore P(\text{getting a face card}) = \frac{16}{52} = \frac{4}{13}.$$

14. Here, $n(S) = 52$.

Let $E =$ event of getting a queen of club or a king of heart.

Then, $n(E) = 2$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{2}{52} = \frac{1}{26}.$$

15. Here, $n(S) = 52$.

There are 26 red cards (including 2 kings) and there are 2 more kings.

Let $E =$ event of getting a red card or a king.

Then, $n(E) = 28$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{28}{52} = \frac{7}{13}.$$

16. Here, $n(S) = 52$.

There are 13 spades (including one ten) and there are 3 more tens.

Let E = event of getting a ten or a spade.

Then, $n(E) = (13 + 3) = 16$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{16}{52} = \frac{4}{13}.$$

17. Here, $n(S) = 52$.

There are 13 cards of diamond (including one king) and there are 3 more kings.

Let E = event of getting a diamond or a king.

Then, $n(E) = (13 + 3) = 16$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{16}{52} = \frac{4}{13}.$$

18. Let S be the sample space.

$$\text{Then, } n(S) = {}^{52}C_2 = \frac{(52 \times 51)}{(2 \times 1)} = 1326.$$

Let E = event of getting 2 kings out of 4.

$$\therefore n(E) = {}^4C_2 = \frac{(4 \times 3)}{(2 \times 1)} = 6.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{1326} = \frac{1}{221}.$$

19. Let S be the sample space.

$$\text{Then, } n(S) = {}^{52}C_2 = \frac{(52 \times 51)}{(2 \times 1)} = 1326.$$

Let E = event of getting 1 spade and 1 heart.

$\therefore n(E)$ = number of ways of choosing 1 spade out of 13 and 1 heart out of 13

$$= ({}^{13}C_1 \times {}^{13}C_1) = (13 \times 13) = 169.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{169}{1326} = \frac{13}{102}.$$

20. Clearly, $n(S) = {}^{52}C_2 = \frac{(52 \times 51)}{2} = 1326$.

Let E_1 = event of getting both red cards,

E_2 = event of getting both kings.

Then, $E_1 \cap E_2$ = event of getting 2 kings of red cards.

$$\therefore n(E_1) = {}^{26}C_2 = \frac{(26 \times 25)}{(2 \times 1)} = 325; n(E_2) = {}^4C_2 = \frac{(4 \times 3)}{(2 \times 1)} = 6;$$

$$n(E_1 \cap E_2) = 2C_2 = 1$$

$$\therefore P(E_1) = \frac{n(E_1)}{n(S)} = \frac{325}{1326}; P(E_2) = \frac{n(E_2)}{n(S)} = \frac{6}{1326};$$

$$P(E_1 \cup E_2) = \frac{1}{1326}$$

$$\therefore P(\text{both red or both kings}) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$= \left(\frac{325}{1326} + \frac{6}{1326} - \frac{1}{1326} \right) = \frac{330}{1326} = \frac{55}{221}.$$

21. Total number of balls = $(6 + 8) = 14$.

Number of white balls = 8.

$$P(\text{drawing a white ball}) = \frac{8}{14} = \frac{4}{7}.$$

22. Total number of balls = $(8 + 7 + 6) = 21$.

Let E = Event that the ball drawn is neither red nor green
= Event that the ball drawn is red.

$$\therefore n(E) = 8.$$

$$\therefore P(E) = \frac{8}{21}.$$

23. Total number of balls = $(4 + 5 + 6) = 15$.

$P(\text{drawing a red ball or a green ball}) = P(\text{red}) + P(\text{green})$

$$= \left(\frac{4}{15} + \frac{5}{15} \right) = \frac{9}{15} = \frac{3}{5}.$$

24. Total number of balls = $(4 + 5 + 3) = 12$.

Let E be the event of drawing 2 red balls.

$$\text{Then, } n(E) = {}^4C_2 = \frac{4 \times 3}{2 \times 1} = 6.$$

$$\text{Also, } n(S) = {}^{12}C_2 = \frac{12 \times 11}{2 \times 1} = 66.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{66} = \frac{1}{11}.$$

25. Total number of balls = $(6 + 4 + 2 + 3) = 15$.

Let E be the event of drawing 2 red balls.

$$\text{Then, } n(E) = {}^6C_2 = \frac{6 \times 5}{2 \times 1} = 15.$$

$$\text{Also, } n(S) = {}^{15}C_2 = \frac{15 \times 14}{2 \times 1} = 105.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{15}{105} = \frac{1}{7}.$$

26. Total number of balls = $(6 + 2 + 4 + 3) = 15$.

Let E be the event of drawing 3 non-yellow balls.

$$\text{Then, } n(E) = {}^{12}C_3 = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220.$$

$$\text{Also, } n(S) = {}^{15}C_3 = \frac{15 \times 14 \times 13}{3 \times 2 \times 1} = 455.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{220}{455} = \frac{44}{91}.$$

27. Total number of marbles = $(6 + 4 + 2 + 3) = 15$.

Let E be the event of drawing 2 blue and 1 yellow marble.

$$\text{Then, } n(E) = ({}^4C_2 \times {}^3C_1) = \frac{4 \times 3}{2 \times 1} \times 3 = 18.$$

$$\text{Also, } n(S) = {}^{15}C_3 = \frac{15 \times 14 \times 13}{3 \times 2 \times 1} = 455.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{18}{455}.$$

28. Total number of marbles = $(6 + 4 + 2 + 3) = 15$.

Let E be the event of drawing 1 green, 2 blue and 1 red marble.

$$\text{Then, } n(E) = ({}^2C_1 \times {}^4C_2 \times {}^6C_1) = 2 \times \frac{4 \times 3}{2 \times 1} \times 6 = 72.$$

$$\text{And, } n(S) = {}^{15}C_4 = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1} = 1365.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{72}{1365} = \frac{24}{455}.$$

29. Total number of marbles = $(6 + 4 + 2 + 3) = 15$.

Let E be the event of drawing 2 marbles such that either both are green or both are yellow.

Then, $n(E) = ({}^2C_1 + {}^3C_2) = (1 + {}^3C_1) = (1 + 3) = 4$. And,
 $n(S) = {}^{15}C_2 = \frac{15 \times 14}{2 \times 1} = 105$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{105}.$$

30. Total number of balls = $(6 + 2 + 4 + 3) = 15$.

Let E be the event of drawing 4 balls such that 2 are red and 2 are green.

Then, $n(E) = ({}^2C_2 \times {}^4C_2) = \left(1 \times \frac{4 \times 3}{2 \times 1}\right) = 6$.

And, $n(S) = {}^{15}C_4 = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1} = 1365$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{1365} = \frac{2}{455}.$$

31. Total number of marbles = $(4 + 5 + 3) = 12$.

Let E be the event of drawing 3 marbles such that none is blue.

Then, $n(E)$ = number of ways of drawing 3 marbles out of 7 = ${}^7C_3 = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$.

And, $n(S) = {}^{12}C_3 = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{35}{220} = \frac{7}{44}.$$

Required probability = $1 - P(E) = \left(1 - \frac{7}{44}\right) = \frac{37}{44}$.

32. Total number of marbles = $(6 + 4 + 2 + 3) = 15$.

Let E be the event of drawing 4 marbles such that none is blue.

Then, $n(E)$ = number of ways of drawing 4 marbles out of 11 non-blue

$$= {}^{11}C_4 = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} = 330.$$

And, $n(S) = {}^{15}C_4 = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1} = 1365$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{330}{1365} = \frac{22}{91}.$$

$$\therefore \text{Required probability} = \left(1 - \frac{22}{91}\right) = \frac{69}{91}.$$

33. Total number of balls = $(6 + 2 + 4 + 3) = 15$.

Let E be the event of drawing 5 balls out of 9 non-blue balls

$$= {}^9C_5 = {}^9C_{(9-5)} = {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126.$$

And, $n(S) = {}^{15}C_5 = \frac{15 \times 14 \times 13 \times 12 \times 11}{5 \times 4 \times 3 \times 2 \times 1} = 3003$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{126}{3003} = \frac{6}{143}.$$

$$\therefore \text{Required probability} = \left(1 - \frac{6}{143}\right) = \frac{137}{143}.$$

34. Total number of balls = $(2 + 3 + 2) = 7$.

Let E be the event of drawing 2 non-blue balls.

Then, $n(E) = {}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10$.

And, $n(S) = {}^7C_2 = \frac{7 \times 6}{2 \times 1} = 21$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{10}{21}.$$

35. Total number of balls = $(10 + 10) = 20$.

Let E be the event of drawing 2 balls of the same colour.

$n(E)$ = number of ways of drawing 2 black balls or 2 white balls

$$n(E) = ({}^{10}C_2 + {}^{10}C_2) = 2 \times {}^{10}C_2 = 2 \times \frac{10 \times 9}{2 \times 1} = 90.$$

$$n(S) = \text{number of ways of drawing 2 balls out of 20} \\ = {}^{20}C_2 = \frac{20 \times 19}{2 \times 1} = 190.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{90}{190} = \frac{9}{19}.$$

36. $P(\text{none is defective}) = \frac{{}^{16}C_2}{{}^{20}C_2} = \left(\frac{16 \times 15}{2 \times 1} \times \frac{2 \times 1}{20 \times 19}\right) = \frac{12}{19}.$

$$P(\text{at least 1 is defective}) = \left(1 - \frac{12}{19}\right) = \frac{7}{19}.$$

37. Let S be the sample space and let E be the event of selecting 2 boys and 1 girl.

Then, $n(S)$ = number of ways of selecting 3 students out of 25 = ${}^{25}C_3 = \frac{25 \times 24 \times 23}{3 \times 2 \times 1} = 2300$.

And, $n(E) = ({}^{15}C_2 \times {}^{10}C_1) = \left(\frac{15 \times 14}{2 \times 1} \times 10\right) = 1050$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{1050}{2300} = \frac{21}{46}.$$

38. $n(S)$ = number of ways of choosing 4 persons out of 9

$$= {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126.$$

$n(E)$ = Number of ways of choosing 2 children out of 4 and 2 persons out of $(3 + 2)$ persons

$$n(E) = ({}^4C_2 \times {}^5C_2) = \left(\frac{4 \times 3}{2 \times 1} \times \frac{5 \times 4}{2 \times 1}\right) = 60.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{60}{126} = \frac{10}{21}.$$

39. Clearly, $n(S) = (6 \times 6) = 36$.

Let E be the event that the sum is a prime number. Then,

$n(E) = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$

$$\therefore n(E) = 15.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}.$$

$$40. P(E) = \frac{30}{100} = \frac{3}{10}, P(H) = \frac{20}{100} = \frac{1}{5} \text{ and } P(E \cap H) = \frac{10}{100} = \frac{1}{10}.$$

$$P(E \text{ or } H) = P(E \cup H)$$

$$= P(E) + P(H) - P(E \cap H) = \left(\frac{3}{10} + \frac{1}{5} - \frac{1}{10} \right) = \frac{4}{10} = \frac{2}{5}.$$

41. Let E_1 = Event that the husband is selected
and E_2 = Event that the wife is selected. Then,

$$P(E_1) = \frac{1}{7} \text{ and } P(E_2) = \frac{1}{5}.$$

$$\therefore P(\bar{E}_1) = \left(1 - \frac{1}{7} \right) = \frac{6}{7} \text{ and } P(\bar{E}_2) = \left(1 - \frac{1}{5} \right) = \frac{4}{5}.$$

$$\therefore \text{Required probability} = P[(A \text{ and not } B) \text{ or } (B \text{ and not } A)]$$

$$= P[(E_1 \cap \bar{E}_2) \text{ or } (E_2 \cap \bar{E}_1)]$$

$$= P[(E_1 \cap \bar{E}_2) + P(E_2 \cap \bar{E}_1)]$$

$$= P(E_1) \cdot P(\bar{E}_2) + P(E_2) \cdot P(\bar{E}_1) = \left(\frac{1}{7} \times \frac{4}{5} \right) + \left(\frac{1}{5} \times \frac{6}{7} \right) = \frac{10}{35} = \frac{2}{7}.$$

42. Let E_1 = Event that A speaks the truth
and E_2 = Event that B speaks the truth. Then,

$$P(E_1) = \frac{75}{100} = \frac{3}{4}, P(E_2) = \frac{80}{100} = \frac{4}{5}, P(\bar{E}_1) = \left(1 - \frac{3}{4} \right)$$

$$= \frac{1}{4}, P(\bar{E}_2) = \left(1 - \frac{4}{5} \right) = \frac{1}{5}.$$

$$P(A \text{ and } B \text{ contradict each other})$$

$$= P[(A \text{ speaks the truth and } B \text{ tells a lie}) \text{ or } (A \text{ tells a lie and } B \text{ speaks the truth})]$$

$$= P[(E_1 \cap \bar{E}_2) \text{ or } (\bar{E}_1 \cap E_2)] = P(E_1 \cap \bar{E}_2) + P(\bar{E}_1 \cap E_2)$$

$$= P(E_1) \cdot P(\bar{E}_2) + P(\bar{E}_1) \cdot P(E_2)$$

$$= \left(\frac{3}{4} \times \frac{1}{5} \right) + \left(\frac{1}{4} \times \frac{4}{5} \right) = \left(\frac{3}{20} + \frac{1}{5} \right) = \frac{7}{20}$$

$$= \left(\frac{7}{20} \times 100 \right) \% = 35\%.$$

43. Let E_1 = Event that A speaks the truth
and E_2 = Event that B speaks the truth.

$$\text{Then, } P(E_1) = \frac{60}{100} = \frac{3}{5}, P(E_2) = \frac{70}{100} = \frac{7}{10}, P(\bar{E}_1)$$

$$= \left(1 - \frac{3}{5} \right) = \frac{2}{5}, P(\bar{E}_2) = \left(1 - \frac{7}{10} \right) = \frac{3}{10}.$$

$$P(A \text{ and } B \text{ say the same thing})$$

$$= P[(A \text{ speaks the truth and } B \text{ speaks the truth}) \text{ or } (A \text{ tells a lie and } B \text{ tells a lie})]$$

$$= P[(E_1 \cap E_2) \text{ or } (\bar{E}_1 \cap \bar{E}_2)] = P(E_1 \cap E_2) + P(\bar{E}_1 \cap \bar{E}_2)$$

$$= P(E_1) \cdot P(E_2) + P(\bar{E}_1) \cdot P(\bar{E}_2)$$

$$= \left(\frac{3}{5} \times \frac{7}{10} \right) + \left(\frac{2}{5} \times \frac{3}{10} \right) = \frac{27}{50} = 0.54.$$

44. Total number of persons = $(3 + 2) = 5$.

$$\therefore n(S) = {}^5C_3 = {}^5C_2 = \frac{5 \times 4}{2 \times 1} = 10.$$

Let E be the event of selecting 3 members having at least 1 woman

Then, $n(E) = n[(1 \text{ woman and } 2 \text{ men}) \text{ or } (2 \text{ women and } 1 \text{ man})]$

$$= n(1 \text{ woman and } 2 \text{ men}) + n(2 \text{ women and } 1 \text{ man})$$

$$= ({}^2C_1 \times {}^3C_2) + ({}^2C_2 \times {}^3C_1) = ({}^2C_1 \times {}^3C_2) + (1 \times {}^3C_1)$$

$$= (2 \times 3) + (1 \times 3) = (6 + 3) = 9.$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{9}{10}.$$

45. Number of blue balls = 3 balls

$$\text{Number of green balls} = 2 \text{ balls}$$

$$\text{Number of red balls} = 5 \text{ balls}$$

$$\text{Total balls in the bag} = 3 + 2 + 5 = 10$$

$$\text{Total possible outcomes} = \text{Selection of 4 balls out of 10}$$

$$\text{balls} = {}^{10}C_4 = \frac{10!}{4!(10-4)!} = \frac{10 \times 9 \times 8 \times 7}{1 \times 2 \times 3 \times 4} = 210$$

$$\text{Favorable outcomes} = (\text{selection of 2 green balls out of 2 balls}) \times (\text{selection of 2 balls out of 3 blue balls})$$

$$= {}^2C_2 \times {}^3C_2$$

$$= 1 \times 3 = 3$$

$$\therefore \text{Required probability} = \frac{\text{Favorable outcomes}}{\text{Total possible outcomes}}$$

$$= \frac{3}{210} = \frac{1}{70}$$

46. Probability of hitting the target:

$$\text{Dev can hit target} \Rightarrow \frac{3}{6} = \frac{1}{2}, \text{ Lakhan can hit target} = \frac{4}{4} = 1$$

$$\text{Pawan can hit target} = \frac{2}{6} = \frac{1}{3}$$

$$\text{Required probability that at least 2 shorts hit target}$$

$$= \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3}$$

$$= \frac{1}{3} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

47. Out of 10 mangoes, 4 mangoes are rotten

$$\therefore \text{Required probability} = \frac{{}^6C_2}{{}^{10}C_2} = \frac{\frac{6!}{2!(6-2)!}}{\frac{10!}{2!(10-2)!}} = \frac{\frac{6!}{2!4!}}{\frac{10!}{2!8!}}$$

$$= \frac{\frac{6 \times 5}{1 \times 2}}{\frac{10 \times 9}{1 \times 2}} = \frac{6 \times 5}{10 \times 9} = \frac{1}{3}$$

48. Number of red balls = 4

$$\text{Number of yellow ball} = 5$$

$$\text{Number of pink ball} = 6$$

$$\text{Total number of balls} = 4 + 5 + 6 = 15$$

$$\text{Total possible outcomes} = \text{selection of 2 balls out of 15}$$

$$\text{balls} = {}^{15}C_2 = \frac{15!}{2!(15-2)!} = \frac{15!}{2!13!} = \frac{15 \times 14}{1 \times 2} = 105$$

Total favourable outcomes = selection of 2 balls out of 4 orange and 6 pink balls.

$${}^{10}C_2 = \frac{10!}{2!(10-2)!} = \frac{10!}{2!8!} = \frac{10 \times 9}{1 \times 2} = 45$$

$$\therefore \text{Required probability} = \frac{45}{105} = \frac{3}{7}$$

49. Number of red balls = 6

Number of yellow ball = 11

Number of pink balls = 5

Total number of balls = 6 + 11 + 5 = 22

$$\begin{aligned} \text{Total possible outcomes} &= n(E) = {}^{22}C_2 = \frac{22!}{2!(22-2)!} = \frac{22!}{2 \times 20!} \\ &= \frac{22 \times 21}{2 \times 1} = 231 \end{aligned}$$

Number of favourable outcomes

$$= n(s) = {}^6C_1 \times {}^{11}C_1 = 6 \times 11 = 66$$

$$\text{Required probability} = \frac{n(E)}{n(S)} = \frac{66}{231} = \frac{2}{7}$$

50. Number of Red balls = 4

Number of Blue balls = 6

Number of Pink balls = 8

Total number of balls = 4 + 6 + 8 = 18

Required probability

$$= \frac{4}{18} \times \frac{11}{20} + \frac{6}{18} \times \frac{11}{20}$$

$$= \frac{11}{20} \left[\left(\frac{4}{18} + \frac{6}{18} \right) \right]$$

$$= \frac{11}{20} \times \frac{10}{18} = \frac{11}{36}$$